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Tendencies toward triadic closure: Field experimental evidence

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Empirical social networks are characterized by a high degree of triadic closure (i.e., transitivity, clustering): network neighbors of the same individual are also likely to be directly connected. It is unknown to what degree this results from dispositions to form such ties (i.e., to close open triangles) per se versus other processes such as homophily and more opportunities for exposure. These mechanisms are difficult to disentangle in many settings. On social media, however, they can be decomposed - and platforms frequently make decisions that depend on these distinct processes. Here, using a field experiment on social media, we randomize the existing network structure that a user faces when they are followed by a target account that we control. We then examine whether the user reciprocates this tie formation. Being randomly assigned to have an existing tie to an account that follows the target user increases tie formation by 35%. Through multiple control conditions, we attribute this effect specifically to a minimal cue that indicates the presence of a potential mutual follower. Theory suggests that triadic closure should be especially likely in open triads of strong ties, and accordingly we find larger effects when the subject has interacted more with the existing follower. These results indicate a substantial role for tendencies toward triadic closure, but one that is substantially smaller than what might be inferred from prior observational studies. Platforms and others may rely on these tendencies in encouraging tie formation, with broader implications for network structure and information diffusion in online networks.

transitivity | field experiment | social network formation

Significance

A ubiquitous feature of social networks is that friends of friends tend to also be friends with each other. But there is surprisingly little evidence that having a mutual connection actually *causes* an increase in the likelihood of forming social ties. This is because mutual connections co-occur with many factors, such as similar interests, that can also contribute to tie formation. By using a randomized social media field experiment, we provide evidence that people are indeed disposed to form ties that close open triangles in their social networks. This result provides unique support for a highly influential sociological theory, and helps resolve an ambiguity about why closed triads are so common.

Social networks affect and reflect numerous social, cultural, and economic processes. Within-organization (1–5), between-organization (6), and population-scale (7–10) networks shape who has access to valuable information and opportunities—with consequences for, e.g., labor market outcomes (11–14) and entrepreneurship (15). Online social networks determine the reach of (mis)information and opinions (16–18), the success of marketing (19), and the adoption of products. Many products and services have network effects, such that network structure then shapes adoption and continued use (20–22). These broad roles have motivated an enduring interest in the structure of social networks, whether at the level of individuals, firms, or regions.

One family of features of individual networks that has been intensively studied is those measuring the extent of triadic closure: Informally, if two individuals share a mutual contact, how likely are they to be directly connected? Triadic closure is a foundational concept in sociology, psychology, and organization studies (13, 23). The extent of triadic closure (or transitivity) is sometimes measured at the level of entire firms, schools (24, 25), or geographic areas (e.g., ref. 7), and other times examined in the ego networks of individuals. Empirical social networks exhibit a striking degree of triadic closure across many domains, such as scientific collaboration networks (26), intrafirm email networks (27), and population-scale communication networks (7, 8). Compared with simple probabilistic models of network formation, real networks typically have an excess of closed triads (28). Considering dynamic networks, people are much more likely to form a tie if they already have mutual contacts (29, 30). For example, Kossinets and Watts (29) find that members of a university community with mutual email contacts are approximately 30 times more likely to begin reciprocally exchanging emails than individuals without such mutual contacts.

This excess of closed triads could result from multiple processes. First, people may have preferences (or, more generally, causally efficacious tendencies) to form ties with their known friends-of-friends or, more generally, others with whom they come to know through mutual network neighbors. This is perhaps the most commonly invoked explanation. We read much of the classic theorizing about triadic closure as about precisely these tendencies. On the balance theory of Heider (23), particularly as elaborated in the context of social networks (e.g., ref. 31), there is a "strain," "imbalance," or "dissonance" associated with some patterns of relationships with others, which could include many open triads, at least once individuals are aware of the other people; this provides a broad psychological account on which people then prefer to resolve this imbalance by forming the triad-closing tie.

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Individuals could vary in their dispositions to form relationships with others they share mutual contacts with, thus explaining persistent heterogeneity in local measures of clustering and closure (32). So one broad explanation of the prevalence of closed triads is that people have a preference for triadic closure per se.

But there are other possible explanations for the prevalence of closed triads in empirical social networks, and it has been difficult to credibly attribute observed levels of triadic closure to these different causes. If individuals have preferences for forming relationships with similar others ("choice homophily"), or if background processes limit their opportunities for encountering dissimilar others (a case of "induced homophily") (29, 33–35)—this readily results in clustered networks (i.e., networks with many closed triads). Adding such homophilous edge formation to random graph models (via, e.g., latent space or stochastic block models) can account for a large portion of the observed triadic closure in many networks [as measured by, e.g., clustering coefficients; (28)]. Prior experiments in online social networks find that randomly assigning people to have (36–38) or see ref. 39 shared traits increases tie formation, and, similarly, having opposite traits increases severing ties (40). In some sense, homophily could thus potentially "explain away" triadic closure, much in the way homophily is confounded with social influence (i.e. peer effects) more generally (41). That is, a question about triadic closure could be understood as about social influence in tie formation: If my alter takes some action (e.g., forming a tie with someone), to what degree does that cause me to take that same action (e.g., forming a tie with the same person)? Even when many traits are observed, fully accounting for their role in ties formation can be complicated by rich interactions, such as with background group structures (34).

We argue that the interpretation of existing empirical evidence is ambiguous. First, as already suggested, it is difficult to distinguish between a causal role for triadic closure versus (choice or induced) homophily: Individuals with mutual contacts may be similar in unobserved ways; these similarities may explain both the existing edges and those that then close the triad. Second, we posit that there are two importantly different ways that an existing open triad (i.e., wedge) can cause triadic closure: The open triad can create new opportunities for tie formation; and, conditional on these opportunities, individuals may differentially form triad-closing ties. People often have more opportunities to form ties with people with whom they have mutual contacts. For example, two employees might meet as one is leaving and the other arrives at the office of a shared colleague. Likewise, recommendation systems, e.g., LinkedIn's "People You May Know", typically recommend ties that would close existing triads (14). This is analogous to differences in opportunities to interact with similar or different others in the literature on homophily (35). Conditional on having an opportunity to form a particular tie, whether that tie would close one or more open triads could further affect whether it forms. So even when an open triad has a causal role in closure, it need not operate through any preference or disposition on the part of individuals.

This multiplicity of processes involved in producing the observed high prevalence of triadic closure is noted in seminal work on consequences of network structure for social and economic outcomes. Granovetter (13), in describing a strong version of the hypothesis that open triads consisting of strong ties are "forbidden" and thus that it is weak ties that are more often part of open triads, noted that this hypothesis is predicted by psychological pressures toward consistency but also consideration of the time commitments involved in stronger social relationships.

Some prior studies have partially accounted for such differences in similarity and opportunities to interact among individuals with and without mutual contacts. In their study of a longitudinal email network, Kossinets and Watts (42) examine rates of tie formation among students with and without a shared social focus (in their case, a common course). We can think of this as improving on an analysis averaging over all pairs of students in that it conditions on a common interest and opportunity to interact. Among students sharing at least one course, there is still a substantial increase in tie formation for pairs with (versus without) one mutual contact—they are approximately 1.9 times more likely to exchange emails. But this is a much smaller relative increase than among those without such a shared social focus (where pairs with one mutual contact are over 100 times more likely to exchange emails).[*] So we might hypothesize that the average 30-times increase in tie formation between students with (versus without) mutual contacts could be largely explained by a combination of the observed social foci (courses) and other, unobserved common social foci (e.g., athletic groups) and shared traits (e.g., national or regional origin). Similarly, in their study of tie formation on Facebook, Sun and Taylor (39) report rates of sending a friend request conditional on visiting another user's profile; on average, users are 1.51 times more likely to send a request if they have mutual contacts (i.e., mutual Facebook friends) than if they have no mutual contacts. However, some or all of this association could be attributed to sharing other common characteristics (e.g., hometown), as users with three such "things in common" are only 33% more likely to send a friend request when they have mutual contacts versus not.[†]

Thus, while the initial association between the presence of one or more open triads and the likelihood of subsequent tie formation is typically very large, we may expect that a more detailed accounting of traits and existing social contexts and ties might reduce or even eliminate such associations. Continuing the analogy with studies of social influence, one could anticipate similar reductions in relative risks of a specific behavior (i.e., tie formation) with and without a mutual contact; for example, when benchmarked against a randomized experiment, Eckles and Bakshy (43) found that naive observational estimates of social influence were 300% too large. If the true relative risks of tie formation in the presence (versus absence) of an open triad were similarly smaller, this would suggest a substantially smaller role for dispositions to close triads than implied by a naive reading of prior work.

In sum, given that many distinct processes can produce the same pattern of frequently closed triads, simply observing a large fraction of closed triads provides limited insight into the underlying social processes. Here, we shed light on this issue by conducting a social media field experiment (44) to directly test for effects of the presence of an open triad on social tie formation. By using a randomized experiment, we are able to rule out homophily and opportunity-based mechanisms and cleanly identify effects of a disposition for closing triads. Our randomized experiment, conducted on Twitter (now called X), enables identification of the effect of an existing open triad on social tie formation in an ecologically valid environment on social media. Specifically, we created a set of human-looking accounts (A users; Fig. 1), used these accounts to follow a set of 2,230 users (B′ users); of these, 214 users (B users) then reciprocated, following back our account that followed them. We then sampled N = 7,311 followers of the B users; these accounts (C users) thus already followed an account that followed one of our accounts (i.e., they had an open triad with one of our accounts). These users were randomly assigned to be followed by the account with whom they had an open triad, or to be followed by an account with whom they did not share a contact (either by using a different account, or by severing the

[*]These quantities and others below from Kossinets and Watts (42) were computed by digitizing the tie formation rates in their Fig. 1*B*.

[†]These quantities from Sun and Taylor (39) were computed using the rounded values reported in their Table 1.

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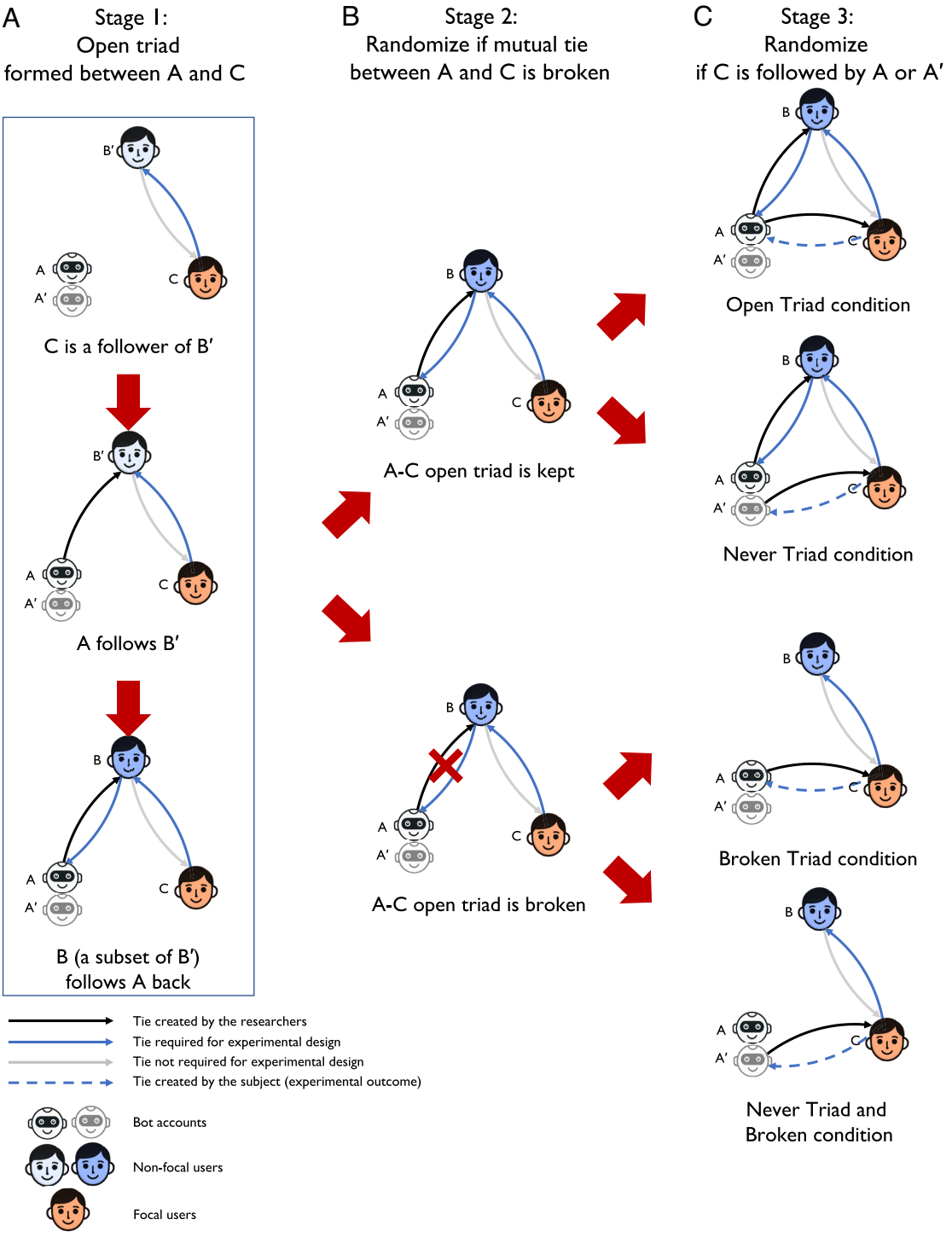


Fig. 1. Field experiment design. (*A*) In Stage 1, we use our human-looking bot accounts (A users) to follow a sample of users producing original posts (B′ users); a subset of those users reciprocally follow our accounts (B users), thereby establishing an open triad between our bot accounts and followers of the accounts that followed our account (C users; subjects in our experiment). C users may or may not have the B user they follow among their followers (i.e., have a reciprocal tie with the B user). (*B*) In Stage 2, we randomly break half of the open triads by removing the A–B ties. (*C*) In Stage 3, we randomize users into four conditions based on being followed by A (the bot account that initially had an open triad with the user) or A′ (the bot account that never had an open triad with the user) and whether the open triad is broken or not. Black lines show the ties that are created by our accounts, blue lines show the ties that have to exist, gray lines show the ties that may exist, and dashed blue lines show the ties that can be created as an outcome of our experiment.

existing tie prior to following them). To ensure users in the control groups would not have any mutual connections with our accounts through users in the treatment group, we immediately severed the connection with every user that reciprocally followed our account.

The essence of our experiment is that each C user was then followed by one of our A (or A′) user accounts, with the presence versus absence of an existing open triad between A and C modulated in multiple possible different ways. Recall that initially, all C users shared a tie (a B user) with one of the A users. We then randomized i) whether we broke the part of the open triad we control (the ties between A and B; Fig. 1*B*) and ii) whether subjects (C users) were followed by the A account they initially shared a

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connection with or the A′ account that they were never connected to (Fig. 1*C*).

This led to the following four experimental conditions. In the "Open Triad" condition, the C user is followed by the A user with whom they share a connection. We then contrast the C user's probability of reciprocating tie formation with the A user in this treatment condition with three different control conditions in which the open triad is absent.

In the "Broken Triad" condition, we break the ties between the B and A users prior to the A user following the C user, thereby removing the open triad (we break the tie by having the A user temporarily block and then unblock the B user, which causes the B user to no longer be following the A user). In the "Never Triad" condition, we follow the C user with a different user with whom the C user had a mutual contact; these A′ users are each paired with an A user and share nearly identical profile information. This control condition allows us to rule out possible effects of a tie being broken in the Broken Triad condition.

In the "Never Triad & Broken" condition, for completeness, we both break the ties between the B and A users, thereby removing the open triad, and follow the C user with a different A user. A statistical reason to include the Never Triad and Never Triad & Broken conditions is that they can be assigned at the level of individual subjects (C users), while the Broken Triad factor can only be assigned at the level of B users (i.e., it is a clustered randomization), which reduces precision. Thus, we expected that including the Never Triad and Never Triad & Broken condition would result in higher power and more precise estimates. If no differences are observed across these three control conditions, it provides reassurance that our controls are not inadvertently inducing unintended effects.

See Fig. 1 and *Methods* for further details of our experimental design. By comparing the probability of the C user reciprocating tie formation with our A user accounts across these 4 conditions, we can identify the causal effect of an open triad (i.e., the presence of a mutual contact) on tie formation.

Results

Rates of reciprocating tie formation by condition are shown in Fig. 2. Following our preregistration (https://aspredicted.org/as4xt.pdf), we first tested whether the reciprocal tie formation rate for the three control conditions where the relevant open triad is absent (Never Triad, Broken Triad, Never Triad & Broken) are significantly different from one another. To do so, we ran a linear regression excluding data from the treatment (Open Triad) condition, predicting whether the C user reciprocated tie formation using dummy variables for being in the Broken Triad and Never Triad & Broken conditions. We conduct a joint significance test of the two dummy coefficients, which we find to be not significantly different from zero [F(2, 5535) = 0.59, P = 0.55]. Using Fisherian randomization (45) inference (with 10,000 permutations) that accounts for the complex stratified and clustered experimental design similarly retains the null hypothesis that the control conditions are equivalent (P_{RI} = 0.63). Thus, we conclude that there is no evidence that tie formation differed significantly between the three control conditions, and thus continue with our planned pooling of the three control conditions to compare with the treatment condition.

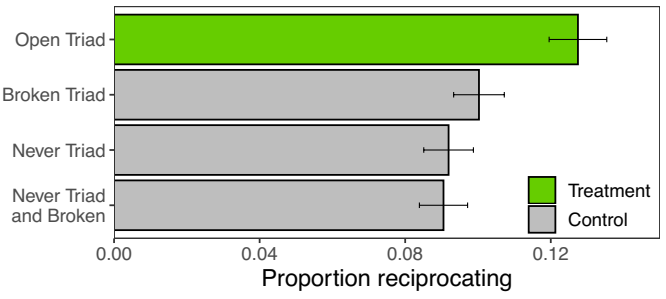


Fig. 2. Probability of reciprocating tie formation in treatment condition versus control conditions. Error bars are ± 1 SE.

We used a linear regression with Huber–White robust "sandwich" SE predicting reciprocating tie formation using a dummy for being in the Open Triad condition. Having an open triad increases the probability of tie formation by 3.3 percentage points (b = 0.033, 95% CI = [0.016, 0.051], SE = 0.009, t(7309) = 3.76, P < 0.001). Given that the reciprocal tie formation rate in the control conditions was 9.4%, this means that the existence of an open triad increases formation of social ties by 35%, CI = [15%, 55%]. A simple calculation multiplying the average number of followers of B users (967) and by this 3.3 percentage point effect translates it into approximately 32 additional ties formed per open triad.

Our results are robust when comparing the treatment with each of the three control conditions separately (b = 0.027, 95% CI = [0.007, 0.048], SE = 0.011, t(3636) = 2.6, P = 0.010 comparing treatment with the Broken Triad condition; b = 0.036, 95% CI = [0.015, 0.056], SE = 0.011, t(3566) = 3.4, P < 0.001 comparing treatment with the Never Triad condition; b = 0.037, 95% CI = [0.017, 0.057], SE = 0.010, t(3649) = 3.6, P < 0.001 comparing treatment with the Never Triad and Broken condition). The results are also robust when including the probability of subjects being bots (46) as a covariate (b = 0.34, P < 0.001), when restricting only to subjects with lower bot likelihood scores (Botometer score < 0.2; N = 3,301: b = 0.031, P = 0.028; Botometer score < 0.3; N = 4,219: b = 0.03, P = 0.014), and when calculating exact P-values using Fisherian randomization inference accounting for the stratified randomization (P_{RI} < 0.001).

We now turn to post hoc analyses of heterogeneous effects. First, from a theoretical perspective, we would expect moderation by the strength of ties to the common node: A stronger link between B and C should make C more inclined to follow our A account. To test this prediction, we ran a linear model in which the treatment dummy is interacted with the number of times the C user mentioned or retweeted the B user (log-transformed due to skew, and then z-scored), a measure of tie strength for the directed tie from C to B. As expected, we find a positive interaction (Table 1, column 1). Thus, the stronger the relationship between B and C, the more having a B-to-A tie increases the probability of forming a C-to-A tie (Fig. 3).

In contrast, additional models do not detect interactions between treatment and log-transformed number of followers or followings in common (a proxy for network similarity between C and B), existence of a reciprocal link between B and C, log-transformed ratio of number of followers to number of followings of B users (a proxy for prominence of B), or log-transformed number of C users per B users (on average 34 C users per each B user with median 18); see Table 1. For completeness, we also note that these models find that mentioning or retweeting the account the users follow were not associated with reciprocal tie formation, while having more common connections with accounts a user follows, reciprocal link between B and C, lower ratio of followers to following of B, and fewer C users per each B are associated with higher reciprocal tie formation rate (Table 1). In other words, the C user is more likely to follow-back our A users when i) there is more of a mutual relationship between B and C (more network overlap between B and C, B follows C), and ii) the B user is less central (lower follower/following ratio, fewer C followers). The positive association between C follow-back and mutual B-C

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Table 1. Separate linear models for relationship between B and C users predicting whether the subject reciprocated tie formation with our account using a Open Triad condition dummy, relationship between C and B users, and the interaction between the two

	(1) Log(#retweets+ #mentions of B by C +1)	(2) #Common followers or followings	(3) Reciprocal link between B and C	(4) Log(#followers / #followings of B + 1)	(5) Log(#C users per each B)
Open triad treatment	0.033*** (0.009)	0.033*** (0.009)	0.032*** (0.009)	0.034*** (0.009)	0.033*** (0.008)
Characteristic of B/ relationship between B and C users	0.002 (0.038)	0.018*** (0.004)	0.021*** (0.003)	−0.010** (0.003)	−0.013** (0.004)
Open triad treatment × Characteristic	0.024* (0.010)	−0.008 (0.009)	−0.017 (0.019)	0.019 (0.010)	−0.01 (0.009)

The relationship variables are z-scored. Shown are regression coefficients with SE in parentheses. (***$P < 0.001$; **$P < 0.01$; *$P < 0.05$).

relationship may reflect greater activity on the part of the C user: If the C user is more active on the platform, that may lead to B and followers of B being more likely to follow C, and to C being more likely to follow both our A account and the accounts followed by B. The negative association between C follow-back and B's centrality likely reflects C's willingness to follow-back low-follower accounts: Our A accounts had relatively few followers, and thus a C account that mostly follows accounts with many followers would be less likely to follow-back our account. The magnitude of these effects is also relevant when contextualizing the size of our treatment effect: Having an additional shared tie causes an increase in follow-back likelihood that is 1.5 to 3 times as large as a 1 SD increase in any of these characteristics.

Finally, we perform post hoc analyses investigating whether the effect of having a shared link is moderated by the characteristics of the C users themselves. Specifically, we examine interactions between our Open Triad treatment and users' ideology [estimated from the ideology of the links they have shared; (47)], number of tweets in the past 2 wk (as a proxy for recent activity level on the platform), probability of being a bot (46), and log-transformed ratio of number of followings to number of followers. We ran separate linear models for each characteristic, predicting whether the subject reciprocally followed our account using an Open Triad condition dummy, the characteristic, and the interaction between the two. None of the interactions were significant ($P > 0.3$ for all), consistent with a relatively homogeneous effect of an open triad. For completeness, we also note that these models reveal variation in baseline rates of reciprocation: Users with characteristics that make them more likely to be bots and who have a larger following to followers ratio are overall less likely to reciprocate, and users who are more active are overall more likely to reciprocate (Table 2). Note that a change of 1 SD in any of these moderators predicts a smaller change in tie formation probability than our Open Triad treatment.

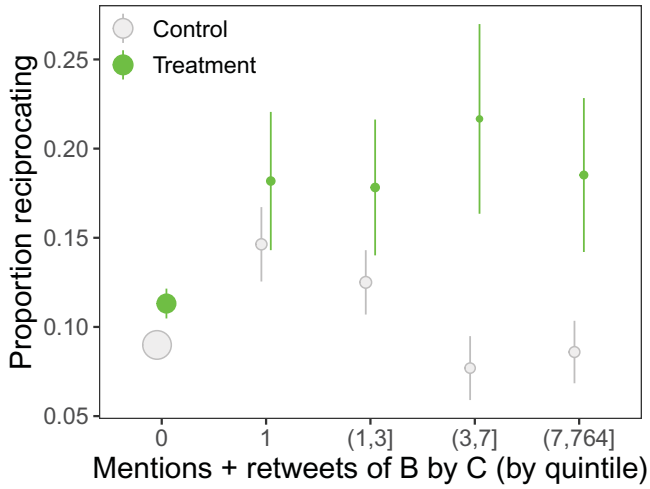


Fig. 3. Heterogeneous effects of treatment by strength of tie from C to B. Probability of reciprocal tie formation in treatment versus control conditions across different values of number of mentions or retweets of B users by C users. Size of dots is proportional to the number of observations in each bin. Error bars are ± 1 SE.

Discussion

Here, we have demonstrated, via a social media field experiment, a large causal effect of an open triad (i.e., the presence of a mutual contact) on tie formation. This result sheds light on the phenomenon of closed triads, where two individuals who share a mutual contact are also connected.

Across many different kinds of social networks, closed triads are much more common than would be expected due to chance under standard random graph models. This prevalence of closed triads is often explained by positing that having a mutual connection with someone makes you more likely to form a connection with them. While arguing that there is no causal role for mutual contacts in new producing triad-closing ties may seem absurd, there is still real uncertainty about how strong any tendency for triadic closure is. In rare cases (e.g., ref. 28), prior work has doubted whether the prevalence of triadic closure cannot simply be explained away by other processes, such as choice or induced homophily; but largely the literature has not explicitly doubted a large role for dispositions toward triadic closure.

From most prior empirical evidence, it is ambiguous as to whether closed triads are actually caused by a disposition toward triadic closure, or simply the result of factors such as homophily (i.e., I may connect to a person with whom I share a connection because we are similar, rather than because we share a connection) or differential opportunities to form connections (48). Such factors could explain observed levels of triadic closure even in the absence of any preferential triadic closure (26). Thus, the widely assumed tendency to close triads has limited direct support in field data. Even taking as given the existence of some such tendency, which is broadly consistent with observations from everyday life, it is unclear how strong such tendencies are.

By randomly assigning whether users have a shared connection, our experiment rules out the possibility of similarity (choice homophily) or differential opportunities (induced homophily) as

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Table 2. Separate linear models for C user characteristics predicting whether the subject reciprocated tie formation with our account using an Open Triad condition dummy, user characteristics, and the interaction between the two

	(1) Ideology	(2) Tweets in past 2 wk	(3) Bot probability	(4) Log[following/(followers+1)]
Open triad treatment	0.033***	0.032***	0.034***	0.033***
	(0.009)	(0.009)	(0.009)	(0.008)
User characteristic	0.005	0.025***	−0.019***	−0.010***
	(0.005)	(0.005)	(0.004)	(0.003)
Open triad treatment × user characteristic	−0.01	0.003	−0.005	−0.000
	(0.010)	(0.010)	(0.008)	(0.007)

Users' characteristics are z-scored. Shown are regression coefficients with SE in parentheses. (***$P < 0.001$; **$P < 0.01$; *$P < 0.05$).

drivers of triadic closure. Thus, our results help to resolve this debate: These results provide unique support for the assumption of a disposition to close triads per se.

On the other hand, our results offer a different quantitative view of the role of tendencies toward triadic closure than a naive reading of prior observational work. In our field experiment, reciprocal tie formation was 35% more likely in the (randomized) presence of an open triad. While substantial, this effect is nearly two orders of magnitude smaller than the 30-times increase found in Kossinets and Watts (42) when averaging over all potential ties. Our estimate is more in line with—although still smaller than—the over 150% increase in tie formation found by Kossinets and Watts (42) among pairs of students when they already share a course, and the 151% lift in tie formation on Facebook found by Sun and Taylor (39) in the presence of at least one mutual contact. Perhaps more comparable to the present results, even conditional on one individual initiating tie formation, Sun and Taylor (39) (Table 1) report data showing that Facebook users accepted friend requests from other users with one or more mutual contacts 156% more often. If one were to interpret these prior results as estimating the causal effect of mutual contacts on reciprocal tie formation, then this would suggest an effect over 3.4 times larger than that revealed by directly randomizing the presence of a mutual contact in our experiment. Compared with naive, unconditional estimates, this is sometimes an orders-of-magnitude-sized downward revision in expectations about the causal effect of an open triad on tie formation. It is a more modest, though still substantial, revision compared with these other estimates that condition on common social context or opportunities to interact—although there may also be differences in the true causal effect across studies due to factors such as the nature of the relationship, the average strength of the mutual tie, and the level of social pressure for mutual tie formation. Generally, however, these reductions are consistent with findings from other behaviors where simple observational estimates of peer influence are often severely inflated (43).

While our experiment identifies tendencies toward triadic closure conditional on opportunities to encounter and interact with others' with mutual ties, it also raises as-yet unanswered questions about the processes underlying this tendency. We mention two that are analogous to reasoning about peer effects in general (49). Information about existing ties can change expectations about the value of forming new ties. In online social networks, i observing that j follows k when i already follows j can function like an endorsement of k by j. Individuals can also have preferences for closing triads. There are a variety of reasons that people might prefer to close triads, such as avoiding dissonance and conflict associated with open triads, the desire to better capitalize on network effects, social norms concerning politeness, or gaining pleasure from co-consuming content. We do not distinguish these expectation and preference interactions directly through the experimental design, which may be a subject for further study. However, we note that given the minimal opportunities for social pressure in our setting, we speculate that our results are primarily driven by changing subjects' expectations about the bot accounts. We thus hypothesize that settings where preferences for closing triads might play a larger role will feature larger tendencies toward triadic closure. This could also be a partial explanation—besides confounding in prior observational studies—of the smaller magnitude of our estimates compared with differences in tie formation rates in social media settings, such as Facebook friend formation, where there may be more of a role for social pressure and norms (39).

Generalizability. How meaningful are the ties that we consider here? To shed light on this question, we used Tweet Analytics to determine the number of impressions that tweets made by the A accounts during Stage 1 (prior to following C users and severing some links) received through the ties they created with the B users. The 26 primary tweets posted by the A accounts received a total of 18,264 impressions and 429 engagements over a period of 40 d. Beyond the paid-for bot followers of these accounts (purchased to make the accounts look more authentic), these accounts had a total of 214 followers (i.e., the B users) during this period. This would translate into an average of 3.28 (median 1.91) impressions per follower and an average of 0.08 (median 0.03) engagements per follower, making the plausible assumption that all impressions come from these B users. This suggests that these social ties are nontrivial vectors of information diffusion.

A potential concern is that the C users are particularly unrepresentative in their tie formation behaviors, e.g., because they follow B users (who are the kind of user willing to reciprocate tie formation with our A accounts). For example, maybe C users (selected because they follow B users) are more likely to be indiscriminate reciprocators compared to the randomly selected group of users followed in Stage 1 (the B users being those who reciprocally followed our account from this group). In contrast to this concern, we find that the reciprocal tie formation rate among C users in the control (9.4%) is extremely similar to the rate among users followed in Stage 1 (9.6%).

Because of the directed nature of many social networks, including in social media (e.g., Instagram, Twitter/X), there are multiple ways local network structure can change that would be understood as triadic closure (32). For example, one case not considered here is when one of two users who already follow the same user might then follow the other. On the other hand, we have studied a case of directed triadic closure consistent with transitivity in the directed relations: B follows A and C follows B, so will C follow A? Platforms clearly regard this as a central case of triadic closure as the ties modulated in our experiment are precisely those that platforms make salient, whether when a user browses to a particular profile or when the platform is showing recommendations of

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who to follow. Further work beyond the context of social media could experimentally investigate similar processes in all types of directed closure or in other settings with undirected ties.

Implications for Social Networks. Beyond demonstrating a tendency for triadic closure, our results have implications for reasoning about origins of social networks more broadly. We have focused on understanding how tendencies toward reciprocal tie formation with others with mutual contacts produces particular local network structures (i.e., closed triangles); this directly affects aggregate patterns in the network (i.e., clustering coefficients). However, we can also see these results as illustrating particular rich-get-richer phenomena in social networks (e.g., ref. 50). Widely used models of network formation feature preferential attachment, whereby nodes are more likely to connect to nodes that already have more connections (e.g., refs. 51 and 52). These results highlight how local network information (i.e., this node is followed by someone I follow) can produce these phenomena, and enable the calibration of these behavioral models.

These results have implications for decisions by platforms, people growing their personal and professional networks, and organizations. Platforms make extensive use of existing ties in recommending new ties. Typically, this involves recommending, and thus causing the formation of, triad closing ties (53), though platforms have at times experimented with making recommendations that result in less clustered [i.e., more structurally diverse networks; (14, 54)]. For individuals, these results demonstrate the indirect consequences of forming a tie: It may result in further, potentially advantageous, tie formation with those connected to the new contact. Finally, organizations, such as firms, increasingly make use of enterprise social media for internal communication, networking, and information sharing. To the extent that organizations want their employees to create more structurally diverse networks (4, 27, 55), they may want to make judicious use of existing affordances that induce triadic closure.

In sum, we show that mutual contacts have a relatively large causal effect on social tie formation in an ecologically valid field experiment on social media. Despite the centrality of triadic closure in theory across several fields, the data in support of it have been almost entirely observational. The present work thus makes theoretical and practical contributions by providing clear, experimental evidence of tendencies for triadic closure, as induced by only a minimal cue that there is a triad to be closed.

Methods

Our experimental design involved three stages, in which we create the open triads and then perform two stages of randomization to determine whether this open triad is present when there is an opportunity for tie formation.

Stage 1: Creating Open Triads. We used a set of four human-looking bot accounts (A users; our accounts; Fig. 4) to follow a sample of 2,230 Twitter users (B′ users). These users were sampled from the stream of users creating original tweet (i.e., non-retweet), were older than 400 d, produced less than 10 tweets per day on average, self-reported being the East Coast of the United States, had less than 10,000 followers, and follow at least half as many users as they have followers. These criteria were designed to limit inclusions of bots and spam accounts, while nonetheless making it likely these accounts might reciprocate in tie formation (which would be unlikely for, e.g., accounts with tens of thousands of followers). Of these sampled accounts, 214 users (B users) reciprocated by following back the account (Fig. 1*A*). As a result, any users who followed any of those 214 B users would now have an open triad with one of our A accounts.

Stages 2 and 3: Tie Formation with and without Open Triads. We randomly selected a sample of N = 7,311 users who followed only one of the B users (C users; subjects in our experiment); see below for more details about the sampling approach. In the language of ref. 32, each of the C users have an out–out wedge, as the C users follow a B user who follows an A user; but there is also an additional (A, B) tie. As described in the introduction and Fig. 1, the C users are either followed by a bot account with whom they share a tie (Open Triad condition) or by a bot account with whom they do not share a tie (Broken Triad, Never Triad, and Never Triad & Broken conditions).

To balance the users' characteristics across experimental conditions and to improve the precision of our causal estimates, we conducted stratified (i.e., blocked) randomization (56). (To avoid confusion with the use of the block feature on Twitter that we use to destroy ties, we refer to these as strata rather than blocks throughout.) First, we created approximately homogeneous strata (with minimum size 4) of B users using log-transformed number of followers and log-transformed followers' average number of accounts followed by the user as

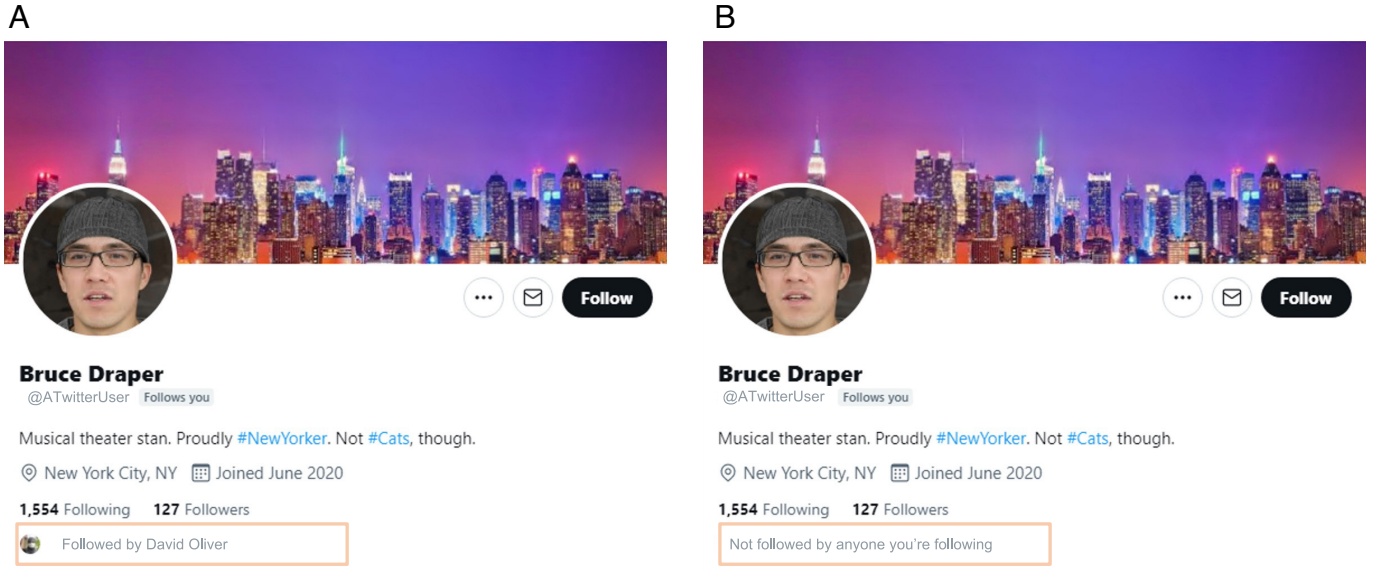


Fig. 4. Sample profiles of human-looking bot accounts. (*A*) In the treatment condition when our human-looking bot account follows the user, the user can see the mutual connection we kept with the user by clicking on the bot's profile. (*B*) In the control condition the user does not see any mutual connection with our bot account on the bot profile since we destroyed that connection before following the user (or that connection did not exist). Orange boxes in the figure are added to emphasize the difference between control and treatment condition.

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the covariates. Within these blocks, we randomly assigned half of these B users either to keep the tie with the A bot or destroy the tie.

We then created approximately homogeneous strata of target subjects (C users) using log-transformed number of accounts followed by the user, log-transformed number of followers, and number of accounts followed by the user divided by number of followers plus one (we chose these variables as they are most related to our potential outcome and log transformed those that had skewed distribution). Within each strata, we then randomly assign C users to be followed by the bot (A) that shares (or shared) an open triad or the paired bot (A′) that never had an open triad; see Fig. 1. We set the minimum size of the blocks such that we can sample users from homogeneous blocks e.g., if we have total of 100,000 users, and want to assign them to two bots each that can follow 1,000 users, we create blocks with minimum size 50 and sample two subjects from each block. We excluded users with more than 10,000 followers (since we expected they were unlikely to reciprocate in following our accounts), as well as users who had more than one mutual tie with our bots to avoid interference between conditions. We followed the subjects over 28 d (roughly 34 follows per A account per day, during the daytime), using eight human-looking bot accounts (Fig. 4). We immediately severed the connection with every user that reciprocally followed our account to avoid any interference (i.e., to prevent those users from becoming mutual ties between our accounts and users in the control conditions).

Our experimental design was approved by and received a waiver of informed consent from the MIT Committee on the Use of Humans as Experimental Subjects (COUHES) protocol #1907910465.

Data, Materials, and Software Availability. All data and scripts necessary to reproduce the results are available in Open Science Framework at https://osf.io/qatj5/ (57).

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